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Burn-In Thermal Control Sets What Aging Sensors Can See

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***Abstract*—**On-chip aging sensors validated under burn-in feed silicon lifecycle management (SLM), yet burn-in benches are specified at the chamber, not at the die the sensor sees. On one FPGA and slack sensor, we compare a stock thermostatic oven with a die-referenced PID oven. The thermostat under-delivers Arrhenius acceleration by 1.65×, through mean die-temperature shortfall rather than ripple, and leaves 1.24× more residual sensor noise. A 214.6-hour PID campaign resolves a sublinear drift whose detection time depends on slow residual wander, and a thermostatic bench would lengthen it by a projected 1.29–1.69×. We propose a reporting checklist.

***Keywords*—**Burn-in, on-chip aging sensor, silicon lifecycle management, FPGA, thermal control, acceleration factor, detectability.

Introduction

On-chip aging sensors are moving from research prototypes into the silicon lifecycle management (SLM) stack, where their readings drive adaptive voltage scaling (AVS), remaining-useful-life estimation, and maintenance decisions [1, 2]. Before a sensor is trusted in the field, it is usually validated under accelerated burn-in: the device is heated and over-volted, and the sensor is expected to report the resulting delay degradation. FPGA aging sensors have been validated this way for over a decade [3].

The standards that govern burn-in, JEDEC JESD22-A108 [4] and MIL-STD-883 Method 1015 [5], set the stress mainly through the *chamber*: ambient temperature, its tolerance, and the applied bias, with junction temperature inferred from them. That a powered device heats itself, so that its junction runs above the chamber, is well known in industrial burn-in; with technology scaling it drove per-device thermal management and junction-temperature control [6, 7]. What has not been examined is how this gap affects the *validation of on-chip sensors*, which sample the die rather than the chamber. The question matters because a sensor's own aging already shifts its calibration [8]: a validation bench that adds its own, unreported stress and noise makes the sensor's behavior harder to separate from the bench's.

Our SBCCI 2026 paper [9] compared a thermostatic (bang-bang) oven and a closed-loop PID oven on the same FPGA and sensor and found that the two benches delivered different stress and different reversible noise. This article asks the practical follow-up: *how long must a burn-in run last before a sensor can see aging, and how much does the bench change that time?* Answering it needs the aging rate the bench induces and the noise it injects, which a 23-hour campaign cannot provide.

The argument is a chain: the bench sets the die-temperature trajectory; its mean fixes the stress delivered and hence the aging signal, its ripple sets the reversible noise, and signal and noise together fix how long a burn-in must run before the sensor sees aging, a limit every SLM function trained on the result inherits. Beyond the conference version, the contributions are:

1. **A source decomposition of the delivered-stress bias.** The 1.65× acceleration-factor gap is set by the mean die-temperature shortfall; thermostat ripple contributes 1.0%, as a second-order expansion predicts. We translate this into a die-level tracking budget.
2. **A 214.6-hour sustained-stress campaign**, first reported here, on a second monitored circuit, in which the sensor resolves a sublinear slack drift after PVT correction.
3. **Detection time, three ways.** A white-noise bound, a long-run-variance estimate, and a sequential test give the burn-in duration needed to see the drift; they differ because the residual wanders slowly and aging is sublinear, and we report all three.
4. **A measured bench penalty on detectability.** The residual noise and decorrelation of both benches are measured on the paired runs; the aging law that converts delivered acceleration into drift rate is the remaining assumption, and we bracket it.
5. **The construction of the PID bench**, from model identification to tuning, with an oven-air comparison at a second setpoint.
6. **A five-item reporting checklist** for any 1-Hz burn-in log.

Instrument, Benches, and Evidence Base

Sensor and DUT. The DUT is a 28 nm Xilinx Artix-7 FPGA on a Nexys 4 DDR board; every campaign below ran on the same board. The instrument is our auto-tuning slack sensor [10]: an MMCM dynamic phase shift advances a sampling clock in fixed steps until an XOR comparator detects that the

This article extends "Quantifying the Effect of Burn-In Thermal Control on Delivered Acceleration Factor and On-Chip Slack-Sensor Fidelity in an FPGA Target," presented at the 39th Symposium on Integrated Circuits and Systems Design (SBCCI 2026) [9]. Digital Object Identifier to be assigned.

monitored path's output has not yet settled, and the number of steps is reported as the path's slack in counts (cnt). The step is $1/(56 f_{\mathrm{VCO}})$: 19.85 ps in the inverter-chain design of the paired runs [9] and 17.86 ps in the bitstream of the long run. The sensor recalibrates autonomously [10]. Die temperature and core voltage V_{CCINT} come from the on-chip XADC, whose absolute offset cancels in the within-run statistics reported here. A programmable supply sets the core voltage, and a supervisory PC logs slack, die temperature and voltage, time-aligned, with oven temperature and supply current where available; time constants below are computed in seconds from each log's timestamps.

Two benches. Both use the same 30-liter laboratory drying oven. The *bang-bang* bench keeps the oven's stock thermostat, the arrangement typical of academic aging labs, with a rate-limited software loop (5 mV steps, at least 2 s apart) that holds the mean supply voltage but not the thermal cycle. The *PID* bench replaces the thermostat with a microcontroller driving a zero-crossing solid-state relay in a 5 s time-proportioning window. An open-loop 50% power step from 25.21 °C identified a first-order-plus-dead-time model with gain 1.56 °C per percent, dead time 150.6 s, and time constant 1307.2 s. Skogestad's SIMC rules [11] gave the starting gains ($K_p = 2.78$, $K_i = 2.31 \times 10^{-3}\,\mathrm{s}^{-1}$). Because the oven overshoots and settles in a way no first-order model captures, the gains were refined on the oven: a lower integral gain and a derivative term ($K_p = 2.78$, $K_i = 1.06 \times 10^{-3}\,\mathrm{s}^{-1}$, $K_d = 5.0\,\mathrm{s}$), plus a soft-start ramp. The stack uses commodity parts: a Cortex-M0+ board, an NTC thermistor, an SSR, and a SCPI-controlled supply. An outer loop moves the oven setpoint in 1 °C steps to hold the die target, using the XADC reading. The thermostatic bench has no such die feedback: its oven setpoint stays where the operator put it. The comparison is therefore between two complete bench architectures as labs deploy them, not between two controllers on an otherwise identical loop.

Is the PID a fair baseline? The thermostatic oven is not a strawman: it is the unmodified commercial chamber. And tuning matters within PID too: at a 100 °C oven-air setpoint, the SIMC starting gains left 1.94 °C steady-state standard deviation and 6.5 °C overshoot, against 0.38 °C and 0.7 °C after refinement.

Evidence base. Table 1 lists the campaigns. The two 23.08-hour runs share every commanded parameter (115 °C die target, 1.3 V on die) and the monitored circuit, a 50-stage inverter chain [9]. They were acquired in separate campaigns [9]: the bang-bang run is the first day of the Week-1 burn-in of [10], and the PID run came months later. The 214.6-hour PID campaign, first reported here, monitors a different circuit, a 16-bit ripple-carry adder, at 125 °C die target (125.05 °C mean, 1.83 °C standard deviation) and 1.4 V. Because the circuits differ, only dimensionless quantities (acceleration-factor and noise ratios) are carried between them. As in [9], stress-delivery statistics cover each whole run, while regression, noise and detectability statistics exclude the first hour as thermal soak.

Table 1. Evidence base. One board, one oven, one sensor design; two monitored circuits (CUT).

Campaign	Bench	CUT, conditions	Used for
23.08 h pair [9]	BB, PID	inverters; 115 °C, 1.3 V	Stress, noise
164.2 h Week 1 [10]	BB	inverters; 115 °C, 1.3 V	Noise check
214.6 h run	PID	adder; 125 °C, 1.4 V	Aging, σ, τ_{int}
1 h each, 100 °C	BB, SIMC, PID	oven air only	2nd setpoint

Delivered Stress: The Mean, Not the Ripple

Burn-in accelerates bias-temperature instability and related mechanisms according to the Arrhenius relation, and for a time-varying die temperature the stress actually delivered is the trajectory average

$$\langle AF \rangle = \left\langle \exp\left[\frac{E_a}{k_B}\left(\frac{1}{T_{\mathrm{use}}} - \frac{1}{T_j(t)}\right)\right]\right\rangle_t. \tag{1}$$

At the same die-temperature target, the thermostatic bench held the die at 107.73 °C on average and the PID bench at 117.04 °C. At $E_a = 0.7\,\mathrm{eV}$ and $T_{\mathrm{use}} = 55$ °C this gives $\langle AF \rangle =$ 31.1 and 51.3, a 1.65× ratio, or 710 versus 1,182 h of equivalent use-condition stress at the mean die temperature over the same 23.08-hour runs [9]. In [9] we related the shortfall to how thermostatic cycling couples to die self-heating; the PID bench's die-referenced setpoint trim contributes as well, and the bang-bang log has no oven-air channel to apportion the two.

A common intuition is that thermostat ripple is what distorts the stress, since the exponential in Eq. (1) weights hot excursions more than cold ones. The measured trajectories contradict it. Ripple raises the thermostatic bench's delivered AF above that of its mean temperature by only 1.0% (PID: 0.2%), although its die-temperature standard deviation is 2.1× larger (2.64 vs. 1.25 °C). This is what a second-order (Jensen) expansion predicts, 1.1%: at these temperatures the exponential is too flat over a few degrees for ripple to matter. The mean shortfall alone predicts a 1.66× ratio, slightly more than the full 1.65×: ripple mildly *offsets* the bias [Fig. 1(a)]. Over $E_a \in [0.5, 0.9]\,\mathrm{eV}$ the ratio spans about 1.4–1.9×, so its direction does not depend on the mechanism assumed.

A lab therefore cannot repair a thermostatic bench by tightening hysteresis; it must know the die's mean temperature, which requires die-level telemetry. A first-order expansion of Eq. (1) gives the mean die-temperature error δ that keeps the delivered AF within a fraction η of nominal,

$$\delta \approx \frac{k_B T_{\mathrm{set}}^2}{E_a} \ln(1+\eta), \tag{2}$$

with T_{set} in kelvin. At 115 °C and 0.7 eV, a ±5%, ±10%, or ±20% AF tolerance allows only 0.9, 1.8, or 3.4 °C. The bang-bang die shortfall of −7.27 °C (an AF deviation of −33%) exceeds even the 20% budget twofold, while the PID die overshoot of 2.04 °C (+12%) is small, stable, and visible [Fig. 1(b)].

The effect is not tied to one setpoint. At 100 °C oven air, the stock thermostat held a 3.38 °C standard deviation, 10.5 °C

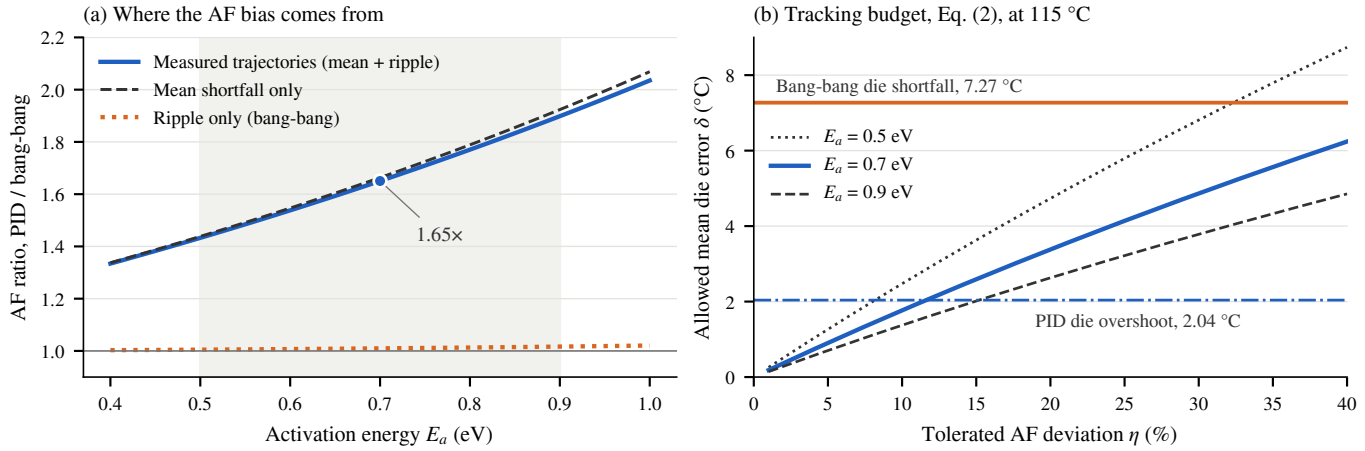


Figure 1. Delivered stress in the 23.08-hour pair, from the measured die trajectories over each whole run. (a) PID-to-bang-bang ratio of delivered acceleration factor versus activation energy. The mean die-temperature shortfall alone reproduces the bias; thermostat ripple contributes 1.0% (shaded: commonly assumed E_a range). (b) Allowed mean die-temperature error from Eq. (2) versus tolerated AF deviation, with each bench's measured mean die error.

peak-to-peak swing and 7.0 °C maximum error over the last 30 minutes, against 0.38, 1.8 and 1.9 °C under PID. These chamber-air figures are what the standards assess. The die-level data show what they leave out: the thermostat's air ripple was the visible symptom, but its mean die shortfall was the consequential error.

Reversible Noise: What the Bench Writes Into the Sensor

The slack a sensor reports mixes a slow, permanent aging term with a fast, reversible PVT term. In the 23.08-hour pair, die temperature and voltage linearly explain 92.1% of slack variance under bang-bang and 59.2% under PID. A lower R^2 under PID is not lower fidelity: the controlled bench simply injects less environmental variance to explain. What matters downstream is the noise left after correction: the residual of the joint temperature–voltage fit is 0.90 cnt under bang-bang and 0.73 cnt under PID [9], a 1.24× higher noise floor for any AVS or alarm calibrated on the thermostatic bench. The supply also oscillated more under bang-bang (6.1 versus 3.6 mV standard deviation) despite compensation, because the die's leakage follows the thermal cycle. These short-term parameters are representative of long runs: with the one-hour moving average removed, the residual of the full 164.2-hour Week-1 log is 0.99 times that of its first day, and the long PID run's first-day residual is 1.01 times that of the whole run.

We tested two candidate mechanisms for the extra noise. Thermal lag between the XADC and the monitored path would make slack hysteretic in temperature; the measured heating-minus-cooling residual gap is 0.11 cnt (12% of the residual), and a dT/dt regressor lowers the residual by 0.6%, so lag is not the mechanism. Nonlinear PVT coupling would leave information after linear correction. On the long PID run, temperature and voltage carry $I(S;T,V) = 1.13$ bits about the slack (3.43 bits of entropy), and a linear correction removes 85% of the short-term coupling; a per-bench comparison of what remains is inconclusive, because integer slack quantization alone reproduces it on the low-noise PID run. The thermostatic excess is therefore broadband variance, which no correction removes.

Detectability: When Can a Sensor See Aging?

A drift is resolved, and it is sublinear. After PVT correction, the 214.6-hour residual drifts at −18.3 m cnt/h on average (95% interval −22.1 to −14.4 with a long-bandwidth Newey–West error), and its correlation with time strengthens from −0.35 before correction to −0.73 after it. The drift is not linear: it slows from −28.3 to −14.8 m cnt/h between the two halves of the run, and a power law t^n with $n = 0.32$ fits far better than a line (ΔBIC = 28 on 2-hour block means), consistent with bias-temperature instability. The total is 3.9 cnt, about 70 ps of lost slack. We call it a drift, not aging, for two reasons: no logged channel can serve as an independent readout control, and the outer loop moved the oven setpoint 49 times (91–99 °C); 12% of these moves coincide with a residual step beyond the 95th percentile at random times (5% expected), a modest source of slow residual wander.

The minimum burn-in time. For a least-squares slope over duration T in noise of variance σ^2 and integrated autocorrelation time $\tau_{\text{int}} = \frac{1}{2} + \sum_k \rho_k$ (in time units), the slope variance is asymptotically $24\,\sigma^2\tau_{\text{int}}/T^3$. Requiring a drift β to exceed three standard errors yields

$$T_{\min} = \left(\frac{216\,\sigma^2\,\tau_{\text{int}}}{\beta^2}\right)^{1/3}. \tag{3}$$

Here β is fitted to the PVT-corrected residual, and σ and τ_{int} describe that residual after its one-hour moving average is removed. With $\sigma = 0.84$ cnt, $\tau_{\text{int}} = 58$ s and the average β, this white-noise estimate gives $T_{\min} = 20$ h. It assumes the residual decorrelates within τ_{int}, and it does not: its long-run variance keeps growing with averaging time, 17-fold from 15-minute to 8-hour blocks. With the 8-hour value in place of $2\sigma^2\tau_{\text{int}}$, $T_{\min} = 89$ h. Conversely, because the drift is fastest early, the cumulative slope crosses 3σ for good after 16 h [Fig. 2(a)], a

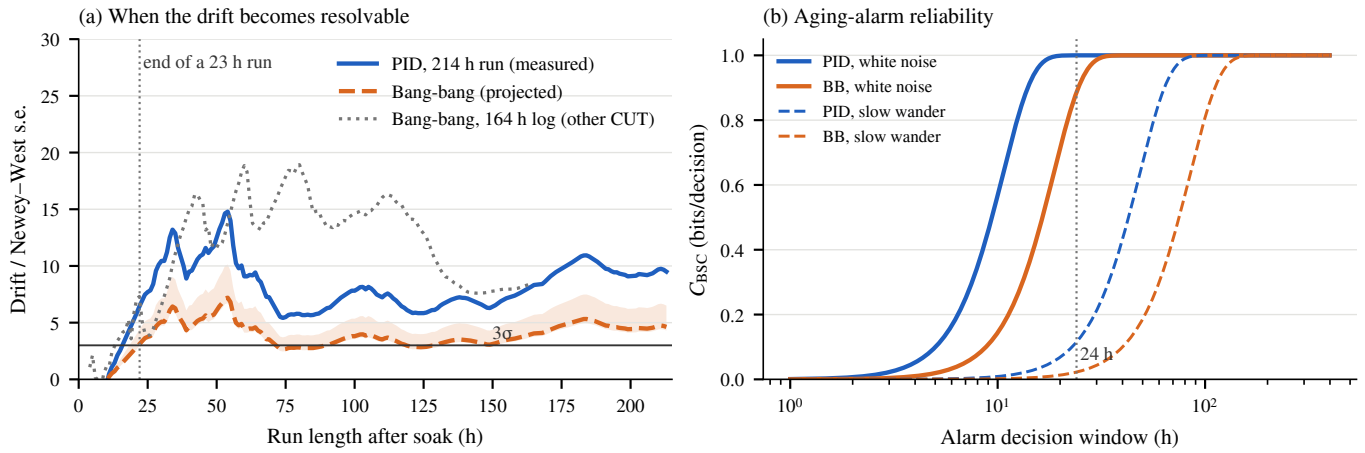


Figure 2. Detectability. (a) Significance of the PVT-corrected slack drift versus run length (cumulative least-squares slope over Newey–West error with bandwidth up to 8 h) on the 214.6-hour PID run, and its projection to a thermostatic bench using the ratios measured on the 23.08-hour pair (band: aging law and τ_{int} ratio, Table 2). Dotted gray: the Week-1 thermostatic log, a different circuit on a then-unstressed device, whose early aging is faster. (b) Capacity of a threshold aging alarm, modeled as a binary symmetric channel, versus decision window: solid, white-noise bound; dashed, with the measured slow residual wander.

Table 2. Burn-in duration (h) to resolve the drift at 3σ, PID long run and projected thermostatic bench (range: aging law and τ_{int} ratio); bottom row: alarm window for 0.99 bits.

Method	Noise / drift model	PID	Thermostatic
Eq. (3)	white, mean rate	20	32 (25–33)
Eq. (3), long-run var.	8 h wander, mean rate	89	144 (114–151)
Sequential test	measured, as observed	16	26 (21–27)
Week-1 log, inverters	measured, unstressed device	–	14 (measured)
Alarm, $C \geq 0.99$	white / 8 h wander	18 / 80	30 / 135

descriptive time rather than a stopping rule. Table 2 collects the three; none is wrong, they answer the question under different assumptions about noise and aging law. The same holds for the 23-hour runs: SBCCI 2026 found nonzero late-window slopes in both at 95% confidence [9], but whether such a slope is aging cannot be decided within a day, which is what the long run settles.

The bench penalty. Eq. (3) shows how the bench enters: through the noise σ and decorrelation τ_{int} it injects, and through the stress it delivers, which sets β. On the paired runs the thermostatic bench raises σ by 1.24×. Compared at a common sampling interval, both inverter-chain residuals are close to white (τ_{int} of 1.3 and 1.2 s), unlike the adder's. The stress enters through the aging law. If drift rate scales with delivered AF, the stress penalty is $1.65^{2/3} = 1.40\times$; if drift follows the fitted power law in equivalent time, $(AFt)^{0.32}$, it is only 1.11×. With the noise penalty of 1.16×, a thermostatic bench would need a projected 1.29–1.62× longer, or 1.69× with the measured τ_{int} ratio (Table 2); after 22 h it would sit at 3.1σ, on the threshold, where the PID run already stood at 6.5σ. This projection carries a noise ratio measured on the inverter chain to the adder, an assumption these data cannot test. The thermostatic Week-1 log itself resolved its drift after 14 h [Fig. 2(a)]: on an unstressed device, early aging is fast enough to show through the thermostat's noise. Detection time depends on the device's aging state as much as on the bench.

Alarm reliability. SLM functions consume decisions, not slopes. We model a threshold aging alarm that compares windows of length w as a binary symmetric channel with crossover probability $p = Q(\mathrm{SNR}_w/2)$, $\mathrm{SNR}_w = \beta w/(\sigma\sqrt{4\tau_{int}/w})$ for two adjacent windows, and report its capacity $C_{BSC} = 1 - H_2(p)$. Under white noise a 24-hour alarm on the PID run errs with probability 0.0% (1.00 bits per decision), the projected thermostatic one with 1.1% (0.91); with the measured slow wander these become 30% and 41% [Fig. 2(b); Table 2]. An alarm window chosen on one bench is mis-set on the other.

From detectability to failure. Detectability does not say how far the circuit is from failing. Our sensor work validated a flip-flop failure catcher at the end of the monitored path that records a real timing violation at the sensor-reported failure point, within clock-jitter margin [10]. Sensor slack thus has a physical zero, and a detected drift becomes a distance to failure under stress. We do not extrapolate a lifetime from one board and one operating point.

What Labs Should Report

The standards behave as designed: they bound the chamber, and the thermostatic bench's air ripple would be flagged by them. What they do not bound is the stress and noise reaching a powered die, which is what a critical-path sensor samples. For sensor-instrumented burn-in we propose the five-item report of Table 3. Each item is computable from a standard 1-Hz log, and together they let a reader judge whether a published aging curve reflects the device or the bench. The two items most often missing in FPGA aging studies, die-level mean temperature and T_{min}, are also the two that change the conclusions most in our data.

Limitations. All data come from one 28 nm board, so bench effects are established within one device, not as population statistics. The two comparisons use different monitored circuits, and the paired runs were separate campaigns at different points in the device's stress history. The thermostatic detectability figures are projections that bracket the

Table 3. Proposed reporting checklist for sensor-instrumented burn-in, with values from this study.

Item	This study
1. Die trajectory: mean, std. dev., error vs. target	BB 107.73 °C, 2.64, −7.27; PID 117.04 °C, 1.25, +2.04
2. Delivered $\langle AF \rangle$ with E_a range	BB 31.1, PID 51.3 at 0.7 eV; ratio 1.43–1.90
3. Post-PVT residual σ	BB 0.90, PID 0.73 cnt (23 h); 0.84 cnt (214.6 h, fast)
4. τ_{int}; long-run variance vs. averaging time	58 s; 17-fold growth, 15 min to 8 h
5. $T_{\min}$ vs. run length, with method	Table 2

aging law and transfer a noise ratio between circuits. No logged channel isolates readout drift. The stress (1.3–1.4 V core, die near 125 °C) exceeds the device's datasheet absolute-maximum core voltage and commercial junction range, so mechanisms beyond field-representative aging may contribute; the findings concern the bench, not the device's lifetime. A matched-duration thermostatic campaign on the same circuit, multi-device replication, and a second technology node are the natural next steps; at 16 nm, published drifts are below 0.1% over 1,000 hours even at elevated stress [12], so the bench-set detection floor will matter even more. Statements about JEDEC and MIL-STD conditions are interpretive and do not claim chamber certification.

Conclusion

On one FPGA board with one sensor, swapping the oven's thermal-control architecture changed the stress delivered by 1.65× and the sensor's residual noise floor by 1.24×. Together these would lengthen the burn-in needed to see a drift by a projected 1.29–1.69×, although a fast-aging device can show its drift through a thermostat's noise within a day, on top of an absolute detection time that already ranges from 16 to 89 h depending on the aging law and the residual's slow wander. The dominant error is the mean die temperature, which chamber-level specifications cannot reveal, not the ripple they police. Sensor-validated burn-in now feeds SLM, so the bench belongs in the measurement chain and should be reported with the same care as the sensor.

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